

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

ANSWER:

1. Detection Performance Comparison

Given:

Model	TP	FP	FN
A – Viola-Jones	80	20	30
B – YOLO	90	15	20

1. DETECTION PERFORMANCE COMPARISON

Two object detection models produce the following results:

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

- (a) Compute precision and recall for both models.
 (b) Compare their performance and state which model is better for object detection.

SOLUTION:

(a) Formulas:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Model A (Viola-Jones)	
Precision _A	$= \frac{80}{80 + 20} = \frac{80}{100} = 0.80 = 80\%$
Recall _A	$= \frac{80}{80 + 30} = \frac{80}{110} = 0.7273 = 72.73\%$

Model B (YOLO)	
Precision _B	$= \frac{90}{90 + 15} = \frac{90}{105} = 0.8571 = 85.71\%$
Recall _B	$= \frac{90}{90 + 20} = \frac{90}{110} = 0.8182 = 81.82\%$

(a) Results Summary:

Model	Precision	Recall
A (Viola-Jones)	80%	72.73%
B (YOLO)	85.71%	81.82%

(b) Comparison:

- Model B (YOLO) has higher precision (85.71%) than Model A (80%).
- Model B (YOLO) has higher recall (81.82%) than Model A (72.73%).
- Higher precision means fewer false positives.
- Higher recall means fewer false negatives.

Final Conclusion:

Model B (YOLO) is better for object detection because it achieves both higher precision (**85.71%**) and higher recall (**81.82%**) than Model A (Viola-Jones). Therefore, **YOLO** provides **better overall detection performance**.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

- (a) Compute average motion magnitude for both methods.
 (b) Compare and evaluate which method captures stronger motion.

ANSWER:

3. OPTICAL FLOW MAGNITUDE COMPARISON

Optical flow magnitudes (in pixels/frame):

Region	Method A (pixels/frame)	Method B (pixels/frame)
R1	5	8
R2	6	10
R3	5	9

- (a) Compute average motion magnitude for both methods.
 (b) Compare and evaluate which method captures stronger motion.

SOLUTION:

(a) Average Motion Magnitude

$$\text{Average Motion Magnitude} = \frac{\text{Sum of magnitudes}}{\text{Number of regions}}$$

$$\text{Average}_A = \frac{5 + 6 + 5}{3} = \frac{16}{3} = 5.33 \text{ pixels/frame}$$

$$\text{Average}_B = \frac{8 + 10 + 9}{3} = \frac{27}{3} = 9 \text{ pixels/frame}$$

(a) Results Summary

Method	Average Motion Magnitude (pixels/frame)
Method A	5.33
Method B	9.00

(b) Comparison and Evaluation

- Method A has an average motion magnitude of 5.33 pixels/frame.
- Method B has an average motion magnitude of 9 pixels/frame.
- A higher optical flow magnitude indicates stronger detected motion.
- Therefore, the method with higher average magnitude captures stronger motion.

Final Conclusion: Method B captures stronger motion because its average motion magnitude (9 pixels/frame) is higher than Method A (5.33 pixels/frame). Therefore, Method B is better for detecting stronger object movement.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

- (a) Compute absolute error for both methods.
 (b) Compare and evaluate which method is more accurate.

ANSWER:

4. DEPTH ESTIMATION COMPARISON (STEREO vs SfM)

Estimated depth values (in meters):

Object	Actual Depth (m)	Stereo Vision (m)	Structure from Motion (SfM) (m)
A	2.0	2.0	2.5
B	3.0	3.0	3.8
C	4.0	4.0	5.0

Actual depths: A = 2 m, B = 3 m, C = 4 m

- (a) Compute absolute error for both methods.
(b) Compare and evaluate which method is more accurate.

SOLUTION:

(a) Absolute Error Calculation

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

Stereo Vision			
Object	Actual (m)	Estimated (m)	Absolute Error (m)
A	2.0	2.0	$ 2.0 - 2.0 = 0.0$
B	3.0	3.0	$ 3.0 - 3.0 = 0.0$
C	4.0	4.0	$ 4.0 - 4.0 = 0.0$
Total Absolute Error (Stereo) = 0.0 m			

Structure from Motion (SfM)			
Object	Actual (m)	Estimated (m)	Absolute Error (m)
A	2.0	2.5	$ 2.5 - 2.0 = 0.5$
B	3.0	3.8	$ 3.8 - 3.0 = 0.8$
C	4.0	5.0	$ 5.0 - 4.0 = 1.0$
Total Absolute Error (SfM) = 2.3 m			

(a) Results Summary

Method	A Error (m)	B Error (m)	C Error (m)	Total Error (m)
Stereo Vision	0.0	0.0	0.0	0.0
SfM	0.5	0.8	1.0	2.3

(b) Comparison and Evaluation

- Stereo Vision has total absolute error = 0.0 m.
- SfM has total absolute error = 2.3 m.
- Lower absolute error indicates more accurate depth estimation.
- Stereo Vision matches the actual depths exactly, while SfM overestimates the depths.

Final Conclusion: Stereo Vision is more accurate because it has zero absolute error (0.0 m), whereas Structure from Motion (SfM) has a total error of 2.3 m. Therefore, Stereo Vision provides better depth estimation for this dataset.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

- (a) Compute accuracy for both systems.
(b) Compare and evaluate which system performs better.

ANSWER:

5. PATTERN RECOGNITION ACCURACY COMPARISON

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

- (a) Compute accuracy for both systems.
(b) Compare and evaluate which system performs better.

SOLUTION:

(a) **Formula:**

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}} \times 100\%$$

Traditional System

$$\begin{aligned}\text{Accuracy} &= \frac{85}{100} \times 100\% \\ &= 0.85 \times 100\% \\ &= \mathbf{85\%}\end{aligned}$$

Deep Learning System

$$\begin{aligned}\text{Accuracy} &= \frac{92}{100} \times 100\% \\ &= 0.92 \times 100\% \\ &= \mathbf{92\%}\end{aligned}$$

(a) **Results Summary:**

System	Accuracy
Traditional	85%
Deep Learning	92%

(b) **Comparison and Evaluation:**

- The Traditional system has an accuracy of **85%**.
- The Deep Learning system has an accuracy of **92%**.
- Deep Learning system has higher accuracy by **7 percentage points**.

Final Conclusion: The **Deep Learning** system performs better than the **Traditional** system because it achieves higher accuracy (92%) compared to 85%. Therefore, **Deep Learning** system is **more effective** for pattern recognition.

6. Precision-Recall Trade-off Analysis

Two models produce:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

- (a) Compute precision and recall for both models.
(b) Compare and evaluate which model is more balanced.

ANSWER:

6 PRECISION-RECALL TRADE-OFF ANALYSIS

Two models produce the following results:

Model	TP (True Positives)	FP (False Positives)	FN (False Negatives)
A	70	30	20
B	85	25	25

(a) **Compute Precision and Recall**

Model A

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{70}{70 + 30} = \frac{70}{100} = 0.70 \\ \text{Precision} &= \mathbf{70\%}\end{aligned}$$

$$\begin{aligned}\text{Recall} &= \frac{TP}{TP + FN} \\ &= \frac{70}{70 + 20} = \frac{70}{90} = 0.778 \\ \text{Recall} &= \mathbf{77.78\%}\end{aligned}$$

Model B

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{85}{85 + 25} = \frac{85}{110} = 0.773 \\ \text{Precision} &= \mathbf{77.27\%}\end{aligned}$$

$$\begin{aligned}\text{Recall} &= \frac{TP}{TP + FN} \\ &= \frac{85}{85 + 25} = \frac{85}{110} = 0.773 \\ \text{Recall} &= \mathbf{77.27\%}\end{aligned}$$

(b) **Compare and Evaluate**

Model	Precision	Recall
A	70%	77.78%
B	77.27%	77.27%

Evaluation



Model B is more balanced because its precision and recall are almost exactly equal.

It also has higher precision while maintaining the same recall level (~77%).

Therefore, **Model B** performs better overall and has a better precision-recall balance.

8. Optical Flow Consistency

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

ANSWER:

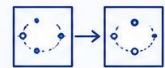
8 OPTICAL FLOW CONSISTENCY

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

Interpretation:

Optical flow values represent the amount of motion detected between consecutive frames.



(a) Compute Average Motion

Method A

$$\text{Average Motion} = \frac{\text{Sum of motions}}{\text{Number of frames}}$$

$$\begin{aligned} &= \frac{4 + 5 + 6}{3} \\ &= \frac{15}{3} \\ &= 5 \end{aligned}$$

Average Motion (A) = 5

Method B

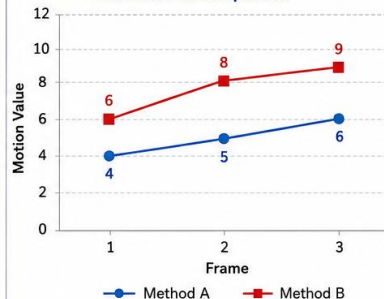
$$\text{Average Motion} = \frac{\text{Sum of motions}}{\text{Number of frames}}$$

$$\begin{aligned} &= \frac{6 + 8 + 9}{3} \\ &= \frac{23}{3} \\ &= 7.67 \end{aligned}$$

Average Motion (B) = 7.67

(b) Compare Consistency and Evaluate

Motion Trend Comparison



Consistency (Variation)

Method A values: 4, 5, 6

$$\begin{aligned} \text{Range} &= \text{Max} - \text{Min} \\ &= 6 - 4 \\ &= 2 \end{aligned}$$

✓ Lower variation → More consistent

Method B values: 6, 8, 9

$$\begin{aligned} \text{Range} &= \text{Max} - \text{Min} \\ &= 9 - 6 \\ &= 3 \end{aligned}$$

✗ Higher variation → Less consistent



Evaluation: Method A is more stable and consistent as it has less variation in motion values.



Summary: Method A → Average Motion = 5, More Consistent (Stable)

Method B → Average Motion = 7.67, Less Consistent (More Variation)

9. Detection Accuracy Comparison

Results:

Model	Correct	Total
A	180	200
B	170	200

(a) Compute accuracy.

(b) Compare and evaluate.

ANSWER:

9 DETECTION ACCURACY COMPARISON

Given Results:

Model	Correct Detections	Total Detections
A	180	200
B	170	200

(a) Compute Accuracy

Model A

$$\text{Accuracy} = \frac{\text{Correct Detections}}{\text{Total Detections}} \times 100\%$$

$$\text{Accuracy(A)} = \frac{180}{200} \times 100\% = 0.90 \times 100\% = 90\%$$

Accuracy of Model A = 90%

Model B

$$\text{Accuracy} = \frac{\text{Correct Detections}}{\text{Total Detections}} \times 100\%$$

$$\text{Accuracy(B)} = \frac{170}{200} \times 100\% = 0.85 \times 100\% = 85\%$$

Accuracy of Model B = 85%

(b) Compare and Evaluate

Model	Accuracy	Percentage
A	0.90	90%
B	0.85	85%



Evaluation:

- Model A has a higher accuracy (90%) compared to Model B (85%).
- This means Model A correctly detects more objects out of the total detections.



Therefore, **Model A** performs better and is more accurate than **Model B**.



Summary:

Model A Accuracy = 90%

Model B Accuracy = 85%



Model A is more accurate.

12. IOU Comparison

IOU values:

Object	A	B
1	0.6	0.8
2	0.7	0.85
3	0.65	0.9

(a) Compute average IOU.

(b) Compare and evaluate detection quality.

ANSWER:

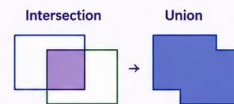
12 IOU COMPARISON

Given IOU Values:

Object	Model A (IOU)	Model B (IOU)
1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

What is IOU?

IOU (Intersection over Union) measures the overlap between the predicted bounding box and the ground truth bounding box. Higher IOU value means better detection quality.



(a) Compute Average IOU

Model A

$$\text{Average IOU} = \frac{\text{Sum of IOU values}}{\text{Number of objects}}$$

$$= \frac{0.60 + 0.70 + 0.65}{3}$$

$$= \frac{1.95}{3}$$

$$= 0.65$$

Average IOU (A) = 0.65

Model B

$$\text{Average IOU} = \frac{\text{Sum of IOU values}}{\text{Number of objects}}$$

$$= \frac{0.80 + 0.85 + 0.90}{3}$$

$$= \frac{2.55}{3}$$

$$= 0.85$$

Average IOU (B) = 0.85

(b) Compare and Evaluate

Model	Average IOU	Detection Quality
A	0.65	Moderate
B	0.85	High



Evaluation:

- Model B has a higher average IOU (0.85) than Model A (0.65).
- This indicates that Model B's predicted bounding boxes overlap more with the ground truth boxes.



Therefore, **Model B** provides **better detection quality** and more accurate localization.



Summary

✓ Average IOU (A) = 0.65

✓ Average IOU (B) = 0.85



Model B has higher IOU values across all objects and a higher average IOU. Hence, **Model B** has **better detection quality** and localization accuracy.



Code of Conduct:

I HARITHI S(192421151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

HARITHI S

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand